DATABASE MANAGEMENT SYSTEMS (10211CS207)

Team Details :

Team Leader:

N Leela prasad	(VTU28813)	(24UECS0238)

Team Members:

NAMES:	**VTU:**	**Registration no:**
E Vishnu	(VTU28235)	(24UECS0550)
B Tanmai Naik	(VTU28506)	(24UECS0443)
Shaik Rehan	(VTU29268)	(24UECS0914)
B Uhalatha	(VTU29924)	(24UECS0415)

TITLE :

HOSPITAL MANAGEMENT SYSTEM

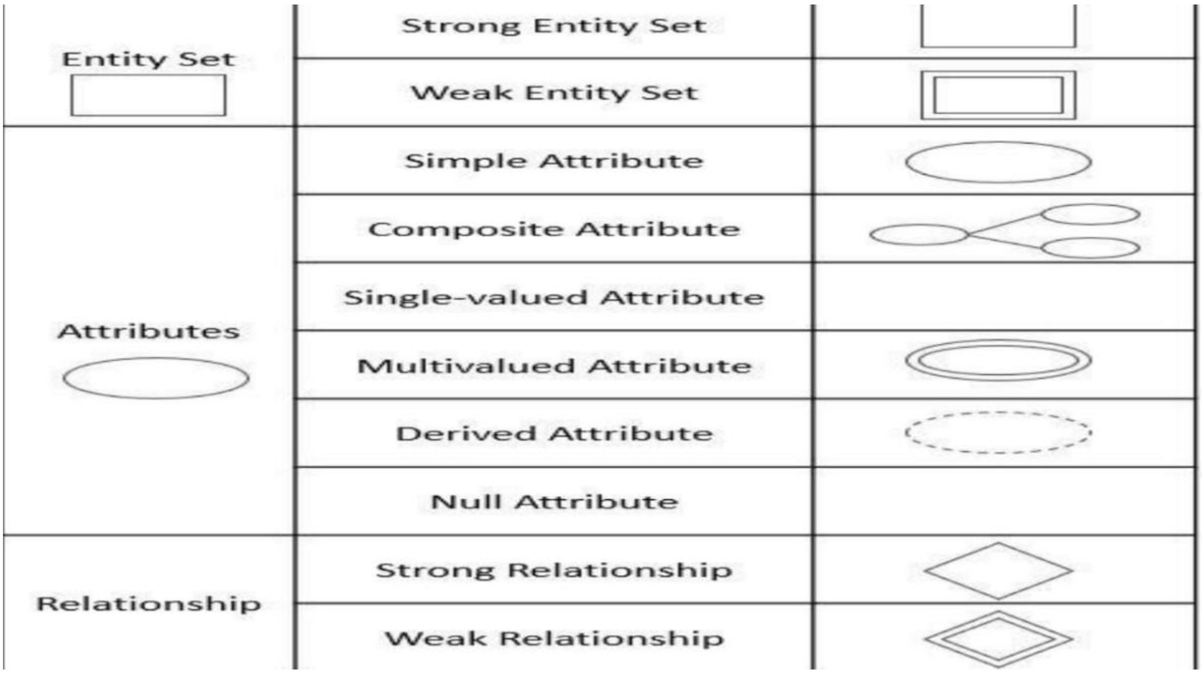
1.ER Diagram:

Aim : To draw the conceptual design for Hospital Management System.

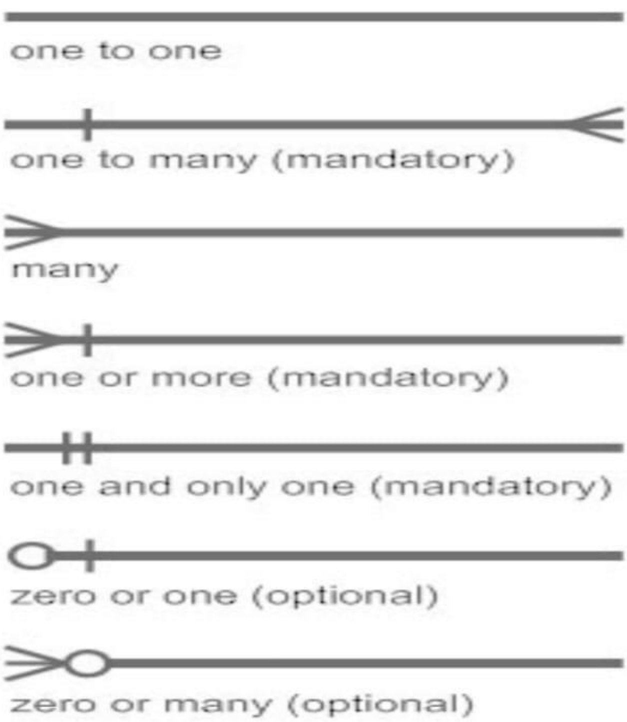
E-R Diagram

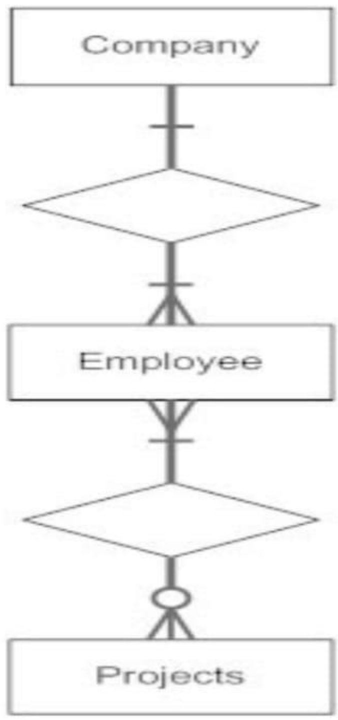
Entity–Relationship model:

- An E-R Diagram (Entity-Relationship Diagram) is a visual tool used to represent the structure of a database — showing entities (objects), their attributes (properties), and relationships.
- It helps design and organize a database clearly before implementation, ensuring proper data connections and reducing redundancy or errors.
- It develops a conceptual design for the database. It also develops a very simple and easy to design view of data



Entity Set	Strong Entity Set	
	Weak Entity Set	
Attributes	Simple Attribute	
	Composite Attribute	
	Single-valued Attribute	
	Multivalued Attribute	
	Derived Attribute	
	Null Attribute	
Relationship	Strong Relationship	
	Weak Relationship	





WEAK ENTITY: An entity that depends on another entity called a weak entity. The weak entity doesn't contain any key attribute of its own. The weak entity is represented by a double rectangle.

ATTRIBUTE: The attribute is used to describe the property of an entity. Eclipse is used to represent an attribute

KEY ATTRIBUTE: The key attribute is used to represent the main characteristics of an entity. It represents a primary key. The key attribute is represented by an ellipse with the text underlined.

COMPOSITE ATTRIBUTE:An attribute that composed of many other attributes is known as a composite attribute. The composite attribute is represented by an ellipse, and those ellipses are connected with an ellipse.

MULTI VALUED ATTRIBUTE : An attribute can have more than one value. These attributes are known as a multivalued attribute. The double oval is used to represent multivalued attribute.

DERIVED ATTRIBUTE: Attributes which are derived from other attributes

ER-MODEL FOR HOSPITAL MANAGEMENT SYSTEM

ENTITIES : Doctor, Patient, Department, Appointment, Treatment, Bill

ATTRIBUTES :

1. Doctor :

- Doctor_ID (PK)
- Name
- Specialization
- Phone
- Department_ID (FK)

2.Patient :

- Patient_ID (PK)
- Name
- Age
- Gender

- Address
- Phone

3. Department

- Department_ID (PK)
- Department_Name

4. Appointment

- Appointment_ID (PK)
- Patient_ID (FK)
- Doctor_ID (FK)
- Appointment_Date
- Reason

5. Treatment

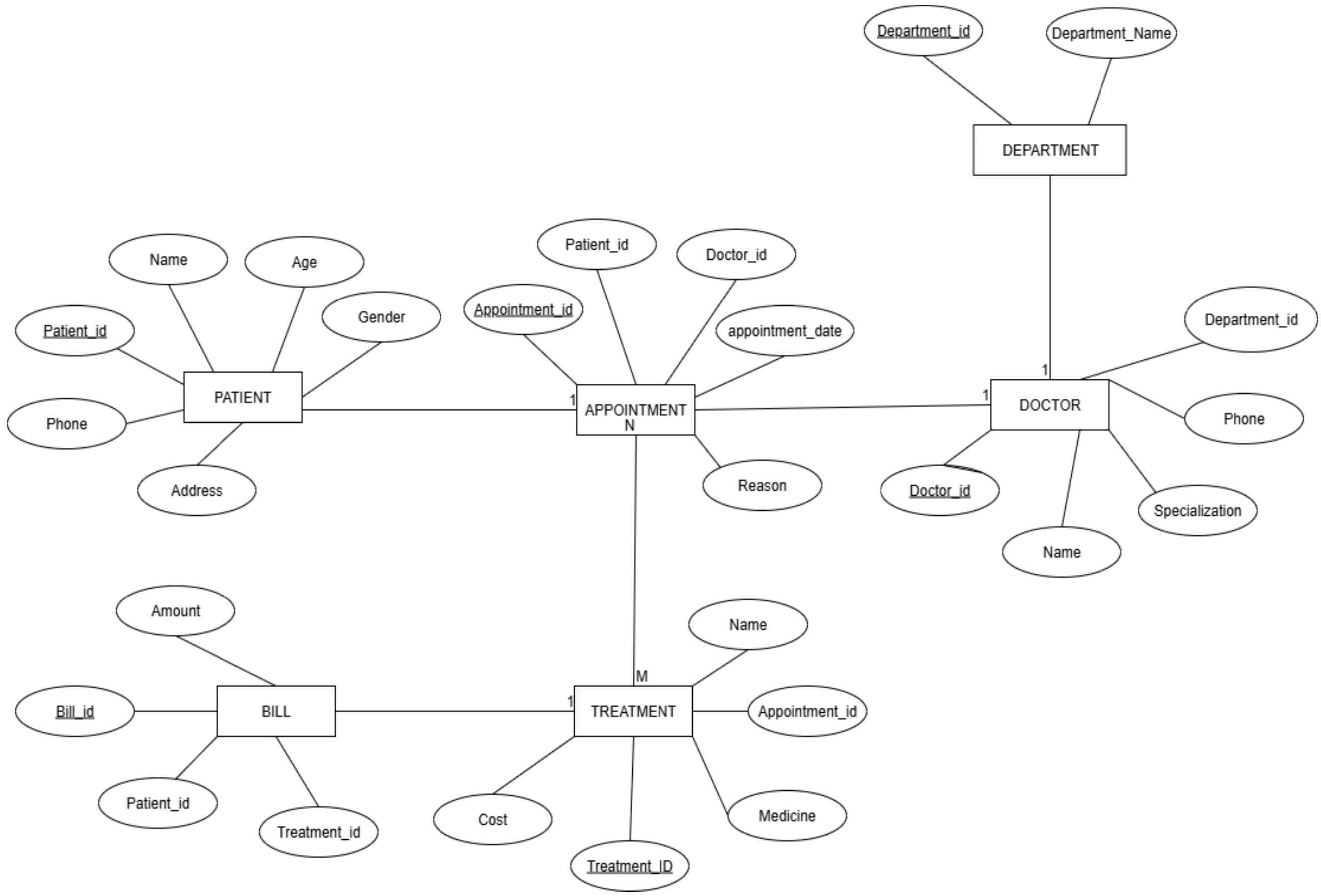
- Treatment_ID (PK)
- Appointment_ID (FK)
- Name
- Medicine
- Cost

6. Bill

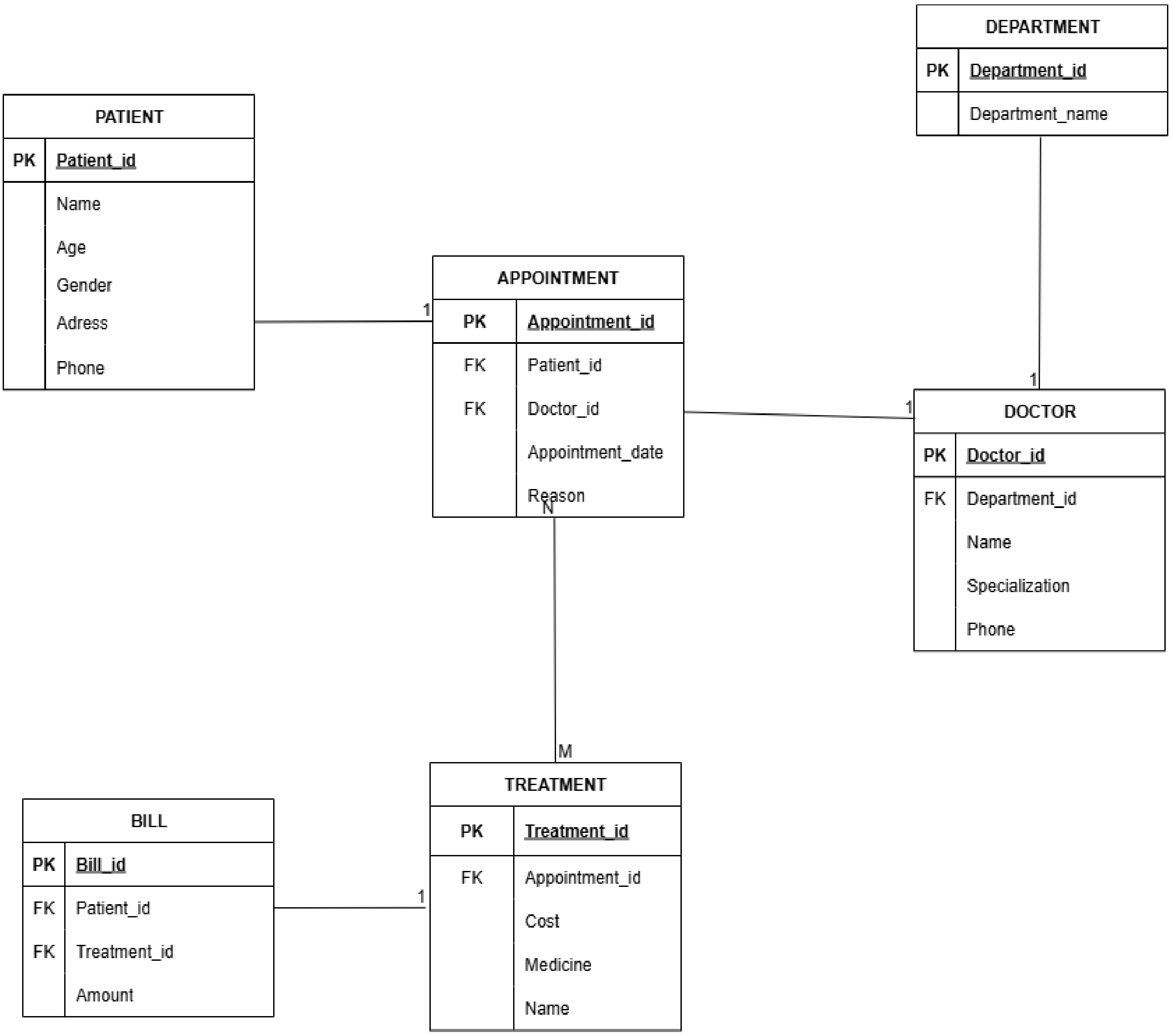
- Bill_ID (PK)
- Patient_ID (FK)
- Treatment_ID (FK)
- Amount

RELATIONS :

Department – Doctor	(1 : M)	One department has many doctors
Doctor – Appointment	(1 : M)	One doctor can have many appointments
Patient – Appointment	(1 : M)	One patient can book many appointments
Appointment – Treatment	(1 : M)	One appointment can include many treatments
Treatment – Bill	(1 : 1)	One treatment has one bill



Department_id
Department_Name
DEPARTMENT
Name
Age
Patient_id
Gender
Phone
PATIENT
Address
Appointment_id
Patient_id
Doctor_id
appointment_date
APPOINTMENT
Reason
Department_id
DOCTOR
Phone
Doctor_id
Specialization
Name
1
1
1
N
M
1
Amount
Bill_id
BILL
Patient_id
Treatment_id
Name
TREATMENT
Appointment_id
Cost
Medicine
Treatment_ID



Result: Thus, the creating er diagram is completed successfully.

SQL QUERIES FOR HOSPITAL MANAGEMENT SYSTEM :

Sql queries are :

CREATE TABLE Department (Department_ID INT PRIMARY KEY, Department_Name VARCHAR(10));

CREATE TABLE Doctor(Doctor_ID INT PRIMARY KEYName VARCHAR(50),Specialization VARCHAR (10),Department_ID INT);

CREATE TABLE Patient (Patient_ID INT PRIMARY KEY, Name VARCHAR(50), Age INT,Gender VARCHAR(10),Address VARCHAR(20), Phone VARCHAR(15));

CREATE TABLE Appointment (Appointment_ID INT PRIMARY KEY,Patient_ID INT,Doctor_ID INT,Appointment_Date DATE,Reason VARCHAR(100));

CREATE TABLE Treatment (Treatment_ID INT PRIMARY KEY,Appointment_ID INT, Name VARCHAR(10),Medicine VARCHAR(10), Cost INT);

CREATE TABLE Bill (Bill_ID INT PRIMARY KEY, Patient_ID INT, Treatment_ID INT, Amount INT);

INSERT INTO Department VALUES

(1, 'Cardiology'),

(2, 'Neurology'),

(3, 'Orthopedics');

INSERT INTO Doctor VALUES

(101, 'Dr. Rahul', 'Cardiologist', 1),

(102, 'Dr. Priya', 'Neurologist', 2),

(103, 'Dr. Arjun', 'Orthopedic', 3);

INSERT INTO Patient VALUES

(201, 'Anita', 30, 'Female', 'Delhi', '9876501234'), (202, 'Rohit', 45, 'Male', 'Mumbai', '9988123456'),

(203, 'Sneha', 28, 'Female', 'Pune', '9123456789');

INSERT INTO Appointment VALUES

(301, 201, 101, '2025-10-01', 'Chest Pain'),

(302, 202, 102, '2025-10-02', 'Headache'),

(303, 203, 103, '2025-10-03', 'Knee Pain');

INSERT INTO Treatment VALUES

(401, 301, 'Heart Checkup', 'Aspirin', 1500.00),

(402, 302, 'MRI Scan', 'Painkillers', 2500.00),

(403, 303, 'Physiotherapy', 'Calcium Tablets', 1800.00);

INSERT INTO Bill VALUES

(501, 201, 401, 1500.00),

(502, 202, 402, 2500.00),

(503, 203, 403, 1800.00);

Patient and Doctor output :

Patient_ID	Name	Age	Gender	Address	Phone
201	Anita	30	Female	Delhi	9876501234
202	Rohit	45	Male	Mumbai	9988123456
203	Sneha	28	Female	Pune	9123456789

Doctor_ID	Name	Specialization	Department_ID
101	Dr. Rahul	Cardiologist	1
102	Dr. Priya	Neurologist	2
103	Dr. Arjun	Orthopedic	3

Appointment and bill output :

Appointment_ID	Patient_ID	Doctor_ID	Appointment_Date	Reason
301	201	101	2025-10-01	Chest Pain
302	202	102	2025-10-02	Headache
303	203	103	2025-10-03	Knee Pain

Bill_ID	Patient_ID	Treatment_ID	Amount
501	201	401	1500
502	202	402	2500
503	203	403	1800

Department and Treatment Output :

Department_ID	Department_Name
1	Cardiology
2	Neurology
3	Orthopedics

Treatment_ID	Appointment_ID	Description	Medicine	Cost
401	301	Heart Checkup	Aspirin	1500
402	302	MRI Scan	Painkillers	2500
403	303	Physiotherapy	Calcium Tablets	1800

NESTED QUERIES FOR HOSPITAL MANAGEMENT SYSTEM :

Patients who have bills greater than the average bill amount

```
SELECT Name
FROM Patient
WHERE Patient_ID IN (
  SELECT Patient_ID
  FROM Bill
  WHERE Amount > (SELECT AVG(Amount) FROM Bill)
);
```

Output

Name
Rohit

Treatments done for Patients older than 40 years

```
SELECT Description
FROM Treatment
WHERE Appointment_ID IN (
  SELECT Appointment_ID
  FROM Appointment
  WHERE Patient_ID IN (
    SELECT Patient_ID
    FROM Patient
    WHERE Age > 40
  )
);
```

Output

Description
MRI Scan

Doctors who have at least one appointment

```sql
SELECT Name
FROM Doctor
WHERE Doctor_ID IN (
  SELECT Doctor_ID
  FROM Appointment
);
```

Output

Name
Dr. Rahul
Dr. Priya
Dr. Arjun

Patient(s) with the Highest Bill Amount

```sql
SELECT Name
FROM Patient
WHERE Patient_ID IN (
  SELECT Patient_ID
  FROM Bill
  WHERE Amount = (SELECT MAX(Amount) FROM Bill)
);
```

Output

Name
Rohit

JOIN QUERIES :

INNER JOIN

Patients with their Doctor Names and Appointment Dates

```sql
SELECT
  P.Name AS Patient_Name,
  D.Name AS Doctor_Name,
  A.Appointment_Date,
  A.Reason
FROM Appointment A
INNER JOIN Patient P ON A.Patient_ID = P.Patient_ID
INNER JOIN Doctor D ON A.Doctor_ID = D.Doctor_ID;
```

Output

Patient_Name	Doctor_Name	Appointment_Date	Reason
Anita	Dr. Rahul	2025-10-01	Chest Pain
Rohit	Dr. Priya	2025-10-02	Headache
Sneha	Dr. Arjun	2025-10-03	Knee Pain

LEFT JOIN

All Patients, Even Those Without Appointments

```sql
SELECT
  P.Name AS Patient_Name,
  A.Appointment_ID,
  A.Appointment_Date
FROM Patient P
LEFT JOIN Appointment A ON P.Patient_ID = A.Patient_ID;
```

Output

Patient_Name	Appointment_ID	Appointment_Date
Anita	301	2025-10-01
Rohit	302	2025-10-02
Sneha	303	2025-10-03

RIGHT JOIN

Departments Even If They Have No Doctors Assigned

```
SELECT
  D.Name AS Doctor_Name,
  Dept.Department_Name
FROM Doctor D
RIGHT JOIN Department Dept ON D.Department_ID = Dept.Department_ID;
```

Output

```
Doctor_Name     Department_Name
Dr. Rahul       Cardiology
Dr. Priya       Neurology
Dr. Arjun       Orthopedics

[Execution complete with exit code 0]
```

FULL OUTER JOIN

All Patients and Their Appointments

```
SELECT
  P.Patient_ID,
  P.Name AS Patient_Name,
  A.Appointment_ID,
  A.Appointment_Date
FROM Patient P
LEFT JOIN Appointment A ON P.Patient_ID = A.Patient_ID

UNION

SELECT
  P.Patient_ID,
  P.Name AS Patient_Name,
  A.Appointment_ID,
  A.Appointment_Date
FROM Patient P
RIGHT JOIN Appointment A ON P.Patient_ID = A.Patient_ID;
```

Output

```
Patient_ID      Patient_Name    Appointment_ID  Appointment_Date
201     Anita   301     2025-10-01
202     Rohit   302     2025-10-02
203     Sneha   303     2025-10-03

[Execution complete with exit code 0]
```

NORMALIZATION :

Normalization in the context of databases refers to the process of organizing data in a database efficiently. The goal is to reduce data redundancy and dependency by organizing fields and table of a database. This helps in minimizing the anomalies that can arise when modifying the data.

There are several normal forms (NF) that define the levels of normalization, with each normal form addressing different types of issues:

First Normal Form (1NF):

- Eliminate duplicate columns from the same table.
- Create a separate table for each group of related data and identify each row with a unique column or set of columns.

Second Normal Form (2NF):

- Meet all the requirements of 1NF.
- Remove partial dependencies—ensure that non-prime attributes are fully functionally dependent on the primary key.

Boyce-Codd Normal Form (BCNF):

- A more stringent form of 3NF.
- For a table to be in BCNF, it must satisfy an additional requirement compared to 3NF, dealing specifically with certain types of functional dependencies.

- In this database we perform normalisation using Griffith university normalisation tool

  Steps to follow for doing normalisation using Griffith normalisation process:

Step1: search for Griffith university normalisation tool in web browser

Step2: After opening the tool enter the attributes of the entity . Make sure to separate the attributes using commas in between them.

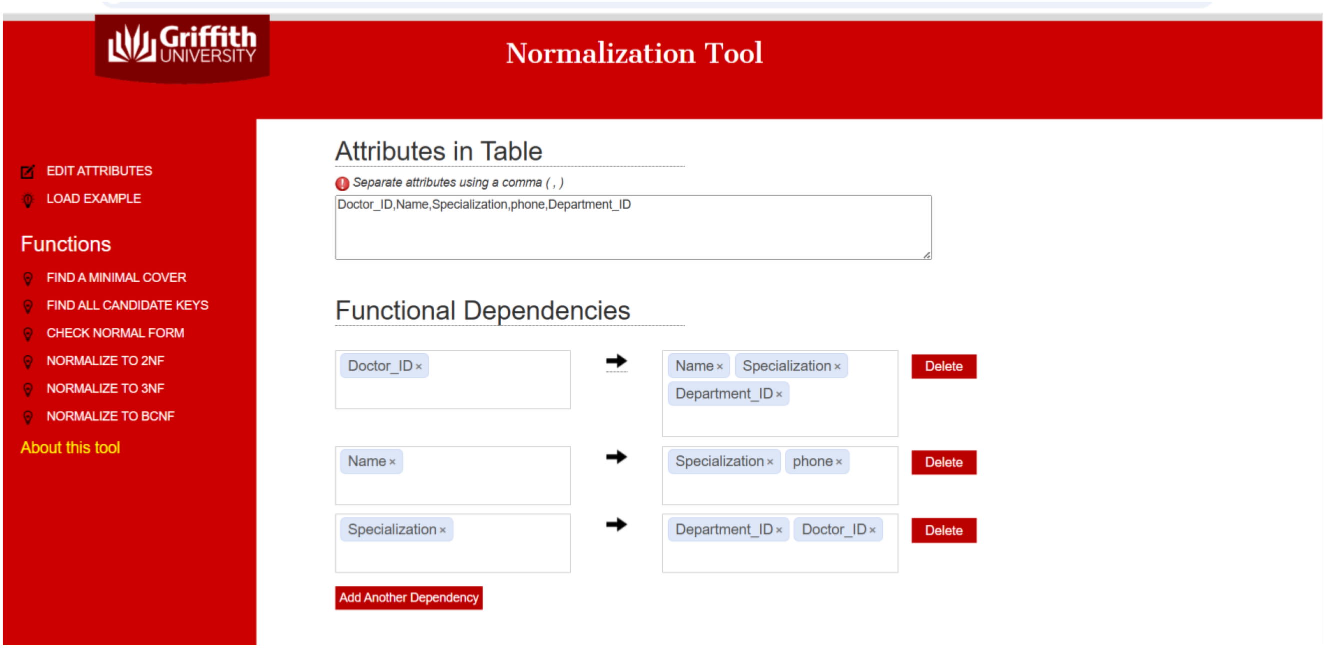
Step3: Add the dependencies of the attributes

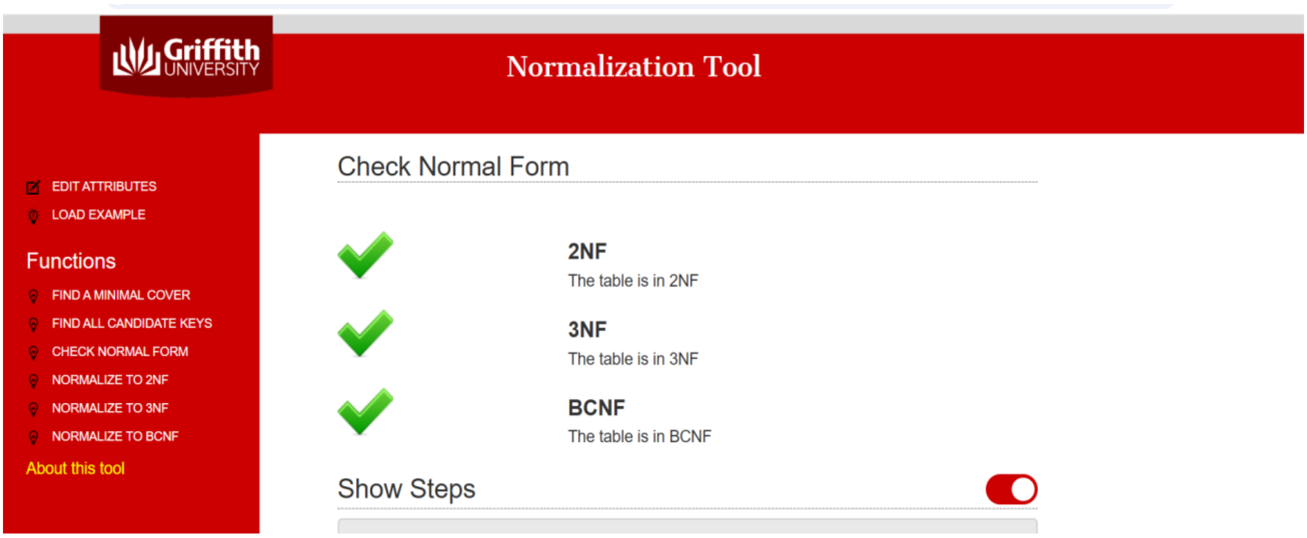
Do as per your entity and add the dependencies as shown in the fig.

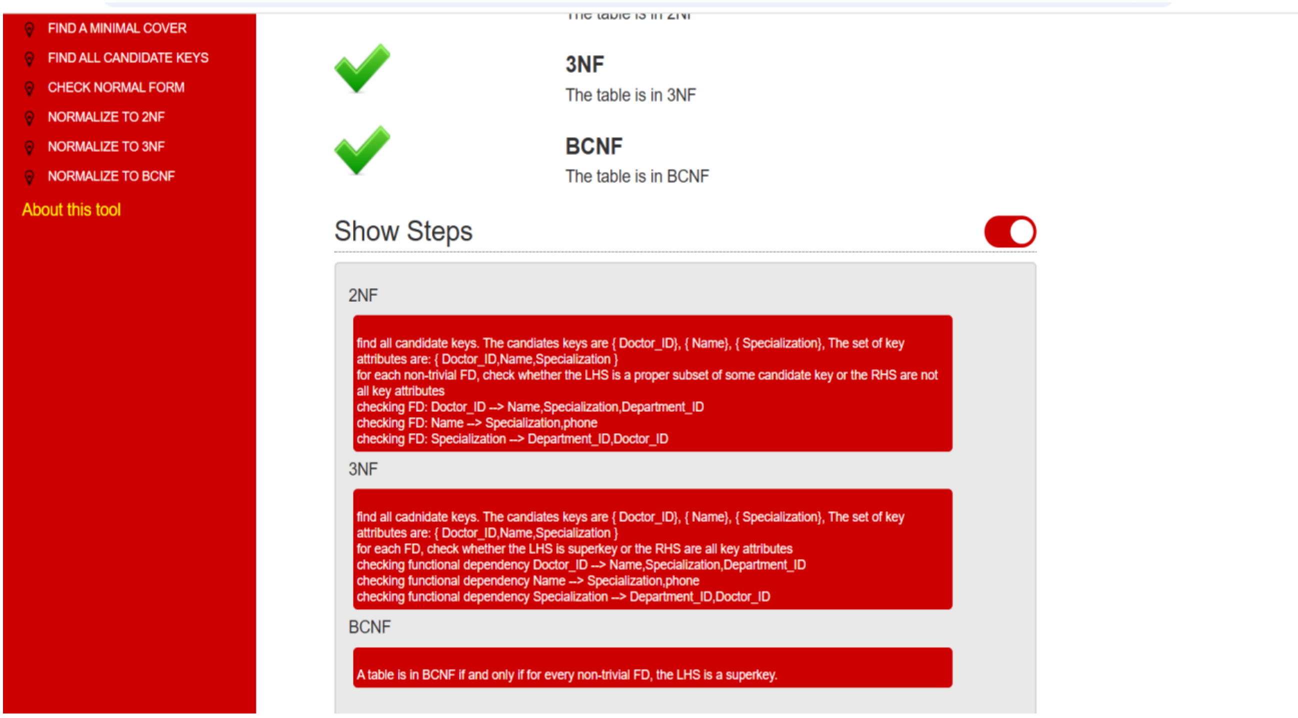
Step4: In the left if the window below functions on check normal form

We will get the screen shown above the normal form of the given attributes is checked (BCNF)

And the following steps are displayed below:







2NF :

1.find all candidate keys. The candiates keys are { Doctor_ID}, { Name}, { Specialization},

The set of key attributes are: { Doctor_ID,Name,Specialization }

2.for each non-trivial FD, check whether the LHS is a proper subset of some candidate

Key or the RHS are not all key attributes

3.checking FD: Doctor_ID --> Name,Specialization,Department_ID

4.checking FD: Name --> Specialization,phone

5.checking FD: Specialization --> Department_ID,Doctor_ID

3NF :

1.find all cadnidate keys. The candiates keys are { Doctor_ID}, { Name}, { Specialization},

The set of key attributes are: { Doctor_ID,Name,Specialization }
2.for each FD, check whether the LHS is superkey or the RHS are all key attributes
3.checking functional dependency Doctor_ID --> Name,Specialization,Department_ID
4.checking functional dependency Name --> Specialization,phone
5.checking functional dependency Specialization --> Department_ID,Doctor_ID

BCNF :

A table is in BCNF if and only if for every non-trivial FD, the LHS is a superkey**.**

Result: Thus, the normalization to 1nf,2nf,3nf, BCNF is completed successfully.

Aim: To implement the document database.

HOSPITAL MANAGEMENT DOCUMENT DATABASE

```
// ------------------------------
// HOSPITAL MANAGEMENT DATABASE
// ------------------------------
use hospitalDB;

// ==============================
// 1 CREATE (Insert Documents)
// ==============================

// ----- Doctor Collection -----
db.doctor.insertMany([
  {
    Doctor_ID: "D001",
    Name: "Dr. Priya",
    Specialization: "Cardiologist",
    Phone: "9876543210",
    Department_ID: "DEP01"
  },
  {
    Doctor_ID: "D002",
    Name: "Dr. Arjun",
    Specialization: "Neurologist",
    Phone: "9876501234",
    Department_ID: "DEP02"
  },
  {
    Doctor_ID: "D003",
    Name: "Dr. Meera",
    Specialization: "Orthopedic Surgeon",
    Phone: "9812345678",
    Department_ID: "DEP03"
  }
]);

// ----- Patient Collection -----
db.patient.insertMany([
  {
    Patient_ID: "P001",
    Name: "Anita",
    Age: 29,
    Gender: "Female",
    Address: "Delhi",
```

```
STDIN
Input for the program ( Optional )

Output:
switched to db hospitalDB
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("68f7245162e7f304d2643e8a"),
                ObjectId("68f7245162e7f304d2643e8b"),
                ObjectId("68f7245162e7f304d2643e8c")
        ]
}
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("68f7245162e7f304d2643e8d"),
                ObjectId("68f7245162e7f304d2643e8e"),
                ObjectId("68f7245162e7f304d2643e8f")
        ]
}
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("68f7245162e7f304d2643e90"),
                ObjectId("68f7245162e7f304d2643e91"),
                ObjectId("68f7245162e7f304d2643e92")
        ]
}
{
        "acknowledged" : true,
```

```
  },
  {
    Patient_ID: "P002",
    Name: "Rohit",
    Age: 35,
    Gender: "Male",
    Address: "Mumbai",
    Phone: "9123400000"
  },
  {
    Patient_ID: "P003",
    Name: "Sneha",
    Age: 24,
    Gender: "Female",
    Address: "Pune",
    Phone: "9876509876"
  }
]);

// ----- Department Collection -----
db.department.insertMany([
  { Department_ID: "DEP01", Department_Name: "Cardiology" },
  { Department_ID: "DEP02", Department_Name: "Neurology" },
  { Department_ID: "DEP03", Department_Name: "Orthopedics" }
]);

// ----- Appointment Collection -----
db.appointment.insertMany([
  {
    Appointment_ID: "A001",
    Patient_ID: "P001",
    Doctor_ID: "D001",
    Appointment_Date: "2025-10-10",
    Reason: "Chest Pain"
  },
  {
    Appointment_ID: "A002",
    Patient_ID: "P002",
    Doctor_ID: "D002",
    Appointment_Date: "2025-10-11",
    Reason: "Headache"
  },
```

```
STDIN
Input for the program ( Optional )

        "insertedIds" : [
                ObjectId("68f7245162e7f304d2643e93"),
                ObjectId("68f7245162e7f304d2643e94"),
                ObjectId("68f7245162e7f304d2643e95")
        ]
}
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("68f7245162e7f304d2643e96"),
                ObjectId("68f7245162e7f304d2643e97"),
                ObjectId("68f7245162e7f304d2643e98")
        ]
}
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("68f7245162e7f304d2643e99"),
                ObjectId("68f7245162e7f304d2643e9a"),
                ObjectId("68f7245162e7f304d2643e9b")
        ]
}
{
        "_id" : ObjectId("68f7245162e7f304d2643e8a"),
        "Doctor_ID" : "D001",
        "Name" : "Dr. Priya",
        "Specialization" : "Cardiologist",
        "Phone" : "9876543210",
        "Department_ID" : "DEP01"
}
```

```
  {
    Appointment_ID: "A003",
    Patient_ID: "P003",
    Doctor_ID: "D003",
    Appointment_Date: "2025-10-12",
    Reason: "Knee Pain"
  }
]);

// ----- Treatment Collection -----
db.treatment.insertMany([
  {
    Treatment_ID: "T001",
    Appointment_ID: "A001",
    Name: "ECG & Medication",
    Medicine: "Aspirin",
    Cost: 1500
  },
  {
    Treatment_ID: "T002",
    Appointment_ID: "A002",
    Name: "MRI Scan",
    Medicine: "Paracetamol",
    Cost: 3000
  },
  {
    Treatment_ID: "T003",
    Appointment_ID: "A003",
    Name: "Physiotherapy",
    Medicine: "Pain Relief Gel",
    Cost: 2000
  }
]);

// ----- Bill Collection -----
db.bill.insertMany([
  {
    Bill_ID: "B001",
    Patient_ID: "P001",
    Treatment_ID: "T001",
    Amount: 1500
  },
```

STDIN

Input for the program (Optional)


```
}
{
        "_id" : ObjectId("68f7245162e7f304d2643e8b"),
        "Doctor_ID" : "D002",
        "Name" : "Dr. Arjun",
        "Specialization" : "Neurologist",
        "Phone" : "9876501234",
        "Department_ID" : "DEP02"
}
{
        "_id" : ObjectId("68f7245162e7f304d2643e8c"),
        "Doctor_ID" : "D003",
        "Name" : "Dr. Meera",
        "Specialization" : "Orthopedic Surgeon",
        "Phone" : "9812345678",
        "Department_ID" : "DEP03"
}
{
        "_id" : ObjectId("68f7245162e7f304d2643e8d"),
        "Patient_ID" : "P001",
        "Name" : "Anita",
        "Age" : 29,
        "Gender" : "Female",
        "Address" : "Delhi",
        "Phone" : "9123456780"
}
{
        "_id" : ObjectId("68f7245162e7f304d2643e8e"),
        "Patient_ID" : "P002",
        "Name" : "Rohit",
```


```
  {
    Bill_ID: "B002",
    Patient_ID: "P002",
    Treatment_ID: "T002",
    Amount: 3000
  },
  {
    Bill_ID: "B003",
    Patient_ID: "P003",
    Treatment_ID: "T003",
    Amount: 2000
  }
]);

// ==============================
// 2 READ (Retrieve Documents)
// ==============================

// Get all doctors
db.doctor.find().pretty();

// Get all patients
db.patient.find().pretty();

// Find doctor by specialization
db.doctor.find({ Specialization: "Cardiologist" });

// Find all appointments for a specific doctor
db.appointment.find({ Doctor_ID: "D003" });

// ==============================
// 3 ADVANCED AGGREGATION (JOINS)
// ==============================

// Patients with their Doctors & Appointments
db.appointment.aggregate([
  {
    $lookup: {
      from: "patient",
      localField: "Patient_ID",
      foreignField: "Patient_ID",
```

STDIN

Input for the program (Optional)


```
        "Age" : 35,
        "Gender" : "Male",
        "Address" : "Mumbai",
        "Phone" : "9123400000"
}
{
        "_id" : ObjectId("68f7245162e7f304d2643e8f"),
        "Patient_ID" : "P003",
        "Name" : "Sneha",
        "Age" : 24,
        "Gender" : "Female",
        "Address" : "Pune",
        "Phone" : "9876509876"
}
{ "_id" : ObjectId("68f7245162e7f304d2643e8a"), "Doctor_ID" : "D001", "Name
{ "_id" : ObjectId("68f7245162e7f304d2643e95"), "Appointment_ID" : "A003",
{ "Appointment_ID" : "A001", "Appointment_Date" : "2025-10-10", "Reason" :
{ "Appointment_ID" : "A002", "Appointment_Date" : "2025-10-11", "Reason" :
{ "Appointment_ID" : "A003", "Appointment_Date" : "2025-10-12", "Reason" :
{ "Bill_ID" : "B001", "Amount" : 1500, "Patient_Info" : [ { "Name" : "Anita
{ "Bill_ID" : "B002", "Amount" : 3000, "Patient_Info" : [ { "Name" : "Rohit
{ "Bill_ID" : "B003", "Amount" : 2000, "Patient_Info" : [ { "Name" : "Sneha
---- Doctors ----
{
        "_id" : ObjectId("68f7245162e7f304d2643e8a"),
        "Doctor_ID" : "D001",
        "Name" : "Dr. Priya",
        "Specialization" : "Cardiologist",
        "Phone" : "9876543210",
```

```
      as: "Patient_Details"
    }
  },
  {
    $lookup: {
      from: "doctor",
      localField: "Doctor_ID",
      foreignField: "Doctor_ID",
      as: "Doctor_Details"
    }
  },
  {
    $project: {
      _id: 0,
      Appointment_ID: 1,
      "Patient_Details.Name": 1,
      "Doctor_Details.Name": 1,
      Appointment_Date: 1,
      Reason: 1
    }
  }
]);

// Show Bill details with patient and treatment info
db.bill.aggregate([
  {
    $lookup: {
      from: "patient",
      localField: "Patient_ID",
      foreignField: "Patient_ID",
      as: "Patient_Info"
    }
  },
  {
    $lookup: {
      from: "treatment",
      localField: "Treatment_ID",
      foreignField: "Treatment_ID",
      as: "Treatment_Info"
    }
  },
```

STDIN

Input for the program (Optional)

```
        "Department_ID" : "DEP01"
}
{
        "_id" : ObjectId("68f7245162e7f304d2643e8b"),
        "Doctor_ID" : "D002",
        "Name" : "Dr. Arjun",
        "Specialization" : "Neurologist",
        "Phone" : "9876501234",
        "Department_ID" : "DEP02"
}
{
        "_id" : ObjectId("68f7245162e7f304d2643e8c"),
        "Doctor_ID" : "D003",
        "Name" : "Dr. Meera",
        "Specialization" : "Orthopedic Surgeon",
        "Phone" : "9812345678",
        "Department_ID" : "DEP03"
}
---- Patients ----
{
        "_id" : ObjectId("68f7245162e7f304d2643e8d"),
        "Patient_ID" : "P001",
        "Name" : "Anita",
        "Age" : 29,
        "Gender" : "Female",
        "Address" : "Delhi",
        "Phone" : "9123456780"
}
{
```

```
  {
    $project: {
      _id: 0,
      Bill_ID: 1,
      "Patient_Info.Name": 1,
      "Treatment_Info.Name": 1,
      Amount: 1
    }
  }
]);

// ==============================
// 4 CHECK ALL DATA
// ==============================
print("---- Doctors ----");
db.doctor.find().pretty();
print("---- Patients ----");
db.patient.find().pretty();
print("---- Departments ----");
db.department.find().pretty();
print("---- Appointments ----");
db.appointment.find().pretty();
print("---- Treatments ----");
db.treatment.find().pretty();
print("---- Bills ----");
db.bill.find().pretty();
```

```
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd1"),
        "Patient_ID" : "P002",
        "Name" : "Rohit",
        "Age" : 35,
        "Gender" : "Male",
        "Address" : "Mumbai",
        "Phone" : "9123400000"
}
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd2"),
        "Patient_ID" : "P003",
        "Name" : "Sneha",
        "Age" : 24,
        "Gender" : "Female",
        "Address" : "Pune",
        "Phone" : "9876509876"
}
---- Departments ----
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd3"),
        "Department_ID" : "DEP01",
        "Department_Name" : "Cardiology"
}
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd4"),
        "Department_ID" : "DEP02",
        "Department_Name" : "Neurology"
}
```

```
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd5"),
        "Department_ID" : "DEP03",
        "Department_Name" : "Orthopedics"
}
---- Appointments ----
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd6"),
        "Appointment_ID" : "A001",
        "Patient_ID" : "P001",
        "Doctor_ID" : "D001",
        "Appointment_Date" : "2025-10-10",
        "Reason" : "Chest Pain"
}
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd7"),
        "Appointment_ID" : "A002",
        "Patient_ID" : "P002",
        "Doctor_ID" : "D002",
        "Appointment_Date" : "2025-10-11",
        "Reason" : "Headache"
}
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd8"),
        "Appointment_ID" : "A003",
        "Patient_ID" : "P003",
        "Doctor_ID" : "D003",
        "Appointment_Date" : "2025-10-12",
        "Reason" : "Knee Pain"
```

---- Treatments ----


```
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dd9"),
        "Treatment_ID" : "T001",
        "Appointment_ID" : "A001",
        "Name" : "ECG & Medication",
        "Medicine" : "Aspirin",
        "Cost" : 1500
}
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dda"),
        "Treatment_ID" : "T002",
        "Appointment_ID" : "A002",
        "Name" : "MRI Scan",
        "Medicine" : "Paracetamol",
        "Cost" : 3000
}
{
        "_id" : ObjectId("68f728985e0ce0dd2c688ddb"),
        "Treatment_ID" : "T003",
        "Appointment_ID" : "A003",
        "Name" : "Physiotherapy",
        "Medicine" : "Pain Relief Gel",
        "Cost" : 2000
}
```

```
---- Bills ----
{
        "_id" : ObjectId("68f728985e0ce0dd2c688ddc"),
        "Bill_ID" : "B001",
        "Patient_ID" : "P001",
        "Treatment_ID" : "T001",
        "Amount" : 1500
}
{
        "_id" : ObjectId("68f728985e0ce0dd2c688ddd"),
        "Bill_ID" : "B002",
        "Patient_ID" : "P002",
        "Treatment_ID" : "T002",
        "Amount" : 3000
}
{
        "_id" : ObjectId("68f728985e0ce0dd2c688dde"),
        "Bill_ID" : "B003",
        "Patient_ID" : "P003",
        "Treatment_ID" : "T003",
        "Amount" : 2000
}
```

Result: Thus implemented the document database has been executed.